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Early warning and recommendation system for undergraduate students at higher education institutions based on machine learning methods

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ABSTRACT

With the rising complexity of student data and the growing need for timely academic interventions, higher education institutions increasingly require intelligent systems to support student success. This research proposes a machine learning-based early warning and recommendation system designed to identify undergraduate students at academic risk and provide personalized strategies to improve their outcomes. A comprehensive dataset containing institutional, academic, and demographic variables was used to train and evaluate fifteen machine learning classifiers, where data from semester N were used to predict academic warning status in semester N + 1, Among them, ensemble and hybrid models such as Boosting with Grid Search, Stacking, and Random Forest outperformed traditional methods in terms of accuracy, recall, and area under the curve (AUC). The best-performing model, Boosting with Grid Search, achieved an accuracy of 89.7, a recall of 0.92, and an AUC of 0.92, demonstrating the strong predictive capability of the proposed approach. Feature selection techniques, including correlation-based and information gain approaches, revealed that academic performance indicators like program average (SCGPA), semester average (grade point average; GPA) and cumulative average (CGPA) were the most predictive features. Beyond early detection, the system incorporates a recommendation module that identifies root causes of academic risk —such as course mismatch and disengagement—and suggests tailored solutions. SHapley Additive exPlanations (SHAP)-based explainability was employed to enhance model transparency, allowing stakeholders to understand and trust the decision-making process. The proposed system offers dual benefits: it improves the efficiency of academic advising while enhancing student retention through data-informed interventions. This work demonstrates the value of integrating predictive analytics with explainable AI (XAI) to support student-centered educational practices.

Subjects Artificial Intelligence, Data Mining and Machine Learning
Keywords Early warning system, Academic risk prediction, Machine learning, SHAP explainability, Feature selection, Ensemble methods, Student retention, Higher education, Academic advising, Predictive analytics

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INTRODUCTION

Higher education institutions (HEIs) now enroll a substantial number of students because the worldwide growth of colleges and universities has become prevalent. The increased

enrollment in higher education institutions has expanded educational opportunities yet it introduced multiple academic difficulties to students and their institutions. The four main academic challenges that result in reduced student retention include early withdrawal rates together with academic caution measures and delayed graduation timelines and shifts between random programs.

Early attrition stands as a serious educational problem because it describes students who depart from institutions before graduation. Educational institutions experience this phenomenon throughout all educational periods including the first year and the end of studies which demonstrates an extensive system-wide challenge in higher education. Various research studies have demonstrated student retention and graduation persistence serve as leading measures for institutional effectiveness yet universities struggle to improve these areas consistently (*Tippetts et al., 2020*; *Sandoval-Palis et al., 2020*; *Khanam, Quraishi & Nazir, 2016*). In 2017 the member countries of the Organization for Economic Co-operation and Development (OECD) showed public universities recorded an average undergraduate dropout rate of about 30% (*Segura, Mello & Hernández, 2022*).

Academic warnings represent a major difficulty which develops from inadequate academic performance or related causes. Traditional academic advising structures fail to support growing student populations with their different academic readiness profiles. Students fail to get necessary guidance promptly because of this omission which prevents them from academic recovery before being subject to dismissal or failure (*Luo, 2020*). More problems emerge because organizations keep changing programs through either unclear decision-making processes or through mismatched connections between student talents and career goals. A significant portion of students in community colleges along with those from lesser-funded schools find it difficult to obtain proper advising services which guide their academic path selection effectively (*Liu, Mishra & Kopko, 2021*).

The academic environment suffers additional complications from students taking longer than expected to complete their programs which academics call "lingering." This situation generates higher expenses for students and educational institutions and reduces educational operational effectiveness. Bachelor's degree attainment takes longer than anticipated for first-time first-year undergraduate students in the United States because their completion rates remain at 40% over a standard four-year period (*Lesinski & Corns, 2018*; *Chalaris et al., 2015*; *Hutt et al., 2019*).

HEIs have adopted conventional academic advising systems to address these problems. During initial parts of semesters advisors handle big student populations while operating under time restrictions. The academic warning detection and communication capabilities on Academic Gate have bettered traditional educational systems yet the advisory process mostly reacts to student issues based on predetermined guidelines. Students receive more procedural approval activities than substantial academic counseling from advisors which restricts their ability to affect student success (*Menke, Duslak & Mcgill, 2020*; *Rahmi et al., 2019*).

Cognitive student support systems are gaining prominence because traditional solutions have shown their inadequacy. Machine learning (ML) delivers a promising response through its ability to build smart scalable systems which process substantial student data

for both pattern recognition and risk prediction and personalized intervention recommendation.

The proposed study outlines the development process of an Early Warning and Recommendation System which uses machine learning algorithms to support undergraduate students in higher education. The primary purpose of this system involves forecasting students who face academic warning risks or potential dropout situations while delivering personalized recommendations for academic success. The system applies descriptive and predictive analytics together with prescriptive analytics to deliver a complete solution for academic counseling.

This research uses multiple student data features including college name, program name, high school grade point average (GPA), school section, university semester details, GPA, cumulative GPA (CGPA), passed credit hours and student status to construct a comprehensive database. The available dataset receives strict preprocessing steps to become ready for building strong machine learning models. The model receives training followed by evaluation based on its ability to properly detect at-risk students along with suitable intervention recommendations.

The research goes beyond traditional early warning system studies by developing a recommendation mechanism that helps students following risk detection. The proposed system achieves two functions through its academic warning detection while performing root cause analysis to deliver custom improvement strategies to students. This system enables students and advisors to develop better decisions because it establishes a connection between predictions and intervention strategies.

The system has two major aims: one includes enhancing academic advising systems and the second ensures reduced faculty workloads and better student retention. The implemented system functions as an advanced data-guided support tool which achieves institutional goals to establish an interactive learning environment that focuses on students. Through explainable artificial intelligence (XAI) methods higher education institutions can build trust in their student success system while also obtaining transparent details about its decision logic, therefore making XAI an important asset for institutions.

RELATED WORKS

Although the existing of several recent works that have already been proposed to address the problem of predicting student retention at an early stage (*Palacios et al., 2021*; *Bañeres et al., 2020*; *Cardona et al., 2020*; *Aulck et al., 2019*; *Hill et al., 2019*; *Isiaka & Abdulsalam, 2019*; *Kumar et al., 2022*; *Zayed, Salman & Hasasneh, 2022*; *Alsayed et al., 2021*; *Bai & Cai, 2018*; *Wang et al., 2019*; *Tsai et al., 2020*; *Kintu, Zhu & Kagambe, 2017*; *Aslam et al., 2021*), most of them did not reach the required accuracy. Some of these recent works are based on machine learning methods (*Palacios et al., 2021*; *Bañeres et al., 2020*; *Cardona et al., 2020*; *Aulck et al., 2019*; *Hill et al., 2019*; *Isiaka & Abdulsalam, 2019*; *Kumar et al., 2022*; *Zayed, Salman & Hasasneh, 2022*; *Alsayed et al., 2021*; *Bai & Cai, 2018*; *Wang et al., 2019*), others are based on statistical models and few of them are based on deep learning models (*Tsai et al., 2020*; *Kintu, Zhu & Kagambe, 2017*; *Aslam et al., 2021*). In particular, the authors in *Palacios et al. (2021)* proposed machine learning algorithms to gain insight into student

retention in higher education. They used various machine learning (ML) algorithms, including K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), Logistic Regression (LR), and Decision Trees (DT), to predict student retention during their first, second, and third academic years. Their models achieved over 80% accuracy in all years. The authors concluded that the RF algorithm was the most effective among the other algorithms considered in this research study.

In *Bañeres et al. (2020)*, researchers focused on creating an early warning system to identify at-risk students in an educational context, covering the entire process from model development to implementation. They trained the system using historical data from the fall semester of 2016, and tested it using data from the most recent semester. Four algorithms, including Naive Bayes (NB), DT, KNN, and SVM were evaluated. By using the NB, it proved to be the most effective classifier based on metrics such as True Positive Rate (TPR) of 93.60% and an F1-score of 92.35%. However, the model showed limitations in accurately identifying at-risk students during initial assessments despite its overall high accuracy within specific courses. Another study (*Cardona et al., 2020*) conducted a systematic review of the literature on the use of ML algorithms to predict student retention in higher education. They found that 19 different ML algorithms were used, with Neural Network (NN), DT, LR, and SVM being the most common techniques. The classification accuracy of the models ranged from 71.59% to 94.00% for NN, 65.38% to 81.36% for DT, 50.18% to 83.07% for LR, and 57.69% to 86.40% for SVM. The DT algorithm showed the highest level of stability and consistency in its results, making it an appropriate option for this specific study subject. In addition, the authors highlighted attributes such as GPA, high school GPA, and financial aid as significant factors in predicting student retention.

In *Aulck et al. (2019)*, the authors modeled first-year student attrition using a dataset of commonly collected data in higher education institutions, such as demographic information, data at the department level, summary data for first-year students, aggregated course data, data on majors, and pre-enrollment data. They used five machine learning models: regularized LR, KNN, RF, SVM, and gradient boosted trees (XGB) to predict student graduation and re-enrollment. The results showed the majority of the models exhibited comparable performance for both predictions. For predicting graduation, LR was the best classifier, followed by the RF. On the other hand, RF achieved the best performance for predicting re-enrollment. Similarly, in *Hill et al. (2019)*, the authors explored two approaches for predicting students who are at risk of failing or leaving of university, namely the RF and the LR algorithms. They found that when students are new to college and have no academic history, the RF algorithm provides a higher level of predictive accuracy. However, when students have completed a semester, the logistic regression algorithm performs better. Other researchers, such as *Isiaka & Abdulsalam (2019)*, have focused on creating an adaptive model for early warning of student dropout using machine learning methods. They tested several algorithms, including KNN, LR, Linear Discriminant Analysis (LDA), Classification and Regression Trees (CART), Gaussian NB, and SVM. The results showed that KNN performed the best and that academic performance played a significant role in predicting dropout.

The researchers in *Kumar et al. (2022)* introduced a recommendation system that combines fuzzy logic and machine learning to predict appropriate academic programs for students. They prepare a dataset with 1,000 records and 21 features, employing the CFS attribute selection algorithm, which narrows the attributes down to 15 from the original 21. The researchers then apply machine learning algorithms, specifically fuzzy SVM, RF, and Classifier version 4.5 (C4.5), to the dataset. The primary goal of the developed model is to predict the optimal academic program for students. Significantly, the results show that the fuzzy SVM algorithm surpasses other algorithms in terms of accuracy within this particular context. Another research study (*Zayed, Salman & Hasasneh, 2022*) has investigated the prediction of undergraduate majors using supervised machine learning techniques such as DT, RF, and SVM. It aimed to improve previous research by incorporating hyperparameter tuning and feature importance analysis. The results showed significant improvements over previous work, particularly with RF achieving an accuracy of 97.70%, compared to 75.00% in previous study (*Alsayed et al., 2021*). The most contributed factors to the predictive model included degree percentage, MBA percentage, and entry test results. Overall, the study improved recommendations for MBA programs by effectively combining academic performance and employment prospects. The authors in *Bai & Cai (2018)*, attempted to improve academic warning systems by replacing manual credit statistics with a mathematical model based on machine learning. They found several indicators to be useful for academic warning, such as family economic status, structure, area, course absenteeism, and credits earned. They used the KNN algorithm. While the authors in *Wang et al. (2019)* introduced a model that detects and predicts student academic problems and facilitates their correction. They collected student data from different departments and applied three machine learning algorithms: DT, Artificial NN, and Bayesian classification algorithms. The findings indicated that the Bayesian algorithm achieved the highest level of accuracy, with an average accuracy rate of 86.00%.

On the other hand, statistical and deep learning models have been used for the early detection of academic warnings and, in particular, of those who intends to drop out of the university. For example, the authors in *Tsai et al. (2020)*, have investigated the predictability of university dropouts using the both a statistical learning technique and a deep neural network approach. They analyzed data from current students and those who had dropped out, utilizing factors such as academic performance, student loan status, attendance, and number of failed courses in both semesters. The results showed a 68.00% accuracy rate using the statistical method and a 77.00% accuracy rate using the deep learning method. Another statistical research work in *Kintu, Zhu & Kagambe (2017)* also emphasizes the significance of learner characteristics and design elements in achieving successful outcomes in a blended learning environment. The study shows gender-neutral performance, while age differences (20–30: 62.00%, 31–39: 67.00%) are statistically significant. Participants demonstrate self-regulation, positive attitudes, and a high level of computer confidence (75.30%). Substantial family and social support (61.00%, 75.00%) further enhances potential effectiveness. Effective management of work-study balance contributes to positive outcomes, despite existing challenges in maintaining satisfaction. Collaborative learning, learner-teacher interaction, and quality of technology all

contribute, supported by periodic in-person support. Noteworthy factors include high intrinsic motivation, satisfaction, and examination outcomes (M = 62.00%, SD = 7.50), which together indicate the promising viability of blended learning. In the context of enhancing early student academic achievement through deep learning, researchers in *Aslam et al. (2021)* used a dataset that included demographic, socio-economic, educational information, and course grades from mathematics and Portuguese language courses. To mitigate the imbalance within the dataset, they used the Synthetic Minority Oversampling Technique (SMOTE). The performance of the model was assessed using different feature combinations and evaluation metrics, including accuracy, precision, recall, and F1-score. The results underscored the significance of the proposed deep learning model in early-stage academic performance prediction. Impressively, this model achieved an accuracy rate of 96.00% for the Portuguese course dataset and 93.00% for mathematics. In addition, substantial precision scores of 99.00% for Portuguese and 94.00% for mathematics were observed. Furthermore, the study highlighted the predictive value of all the features in determining student academic performance.

These studies have used various machine learning algorithms, including KNN, SVM, RF, LR, D T, NN, XGB, LDA, CART, Gaussian NB, Bayesian Classification, and Regression Analysis, to predict student retention and academic performance in higher education. These attempts have shown that different algorithms have different levels of accuracy, with some proving to be more effective than others. For instance, RF emerged as the most successful algorithm in one study, while KNN produced the best results in another study. Moreover, these studies have identified additional factors, such as academic performance, family economic status, and student loan status, as important contributors to predicting student retention and academic performance. However, a notable drawback of the current approaches is the limited interpretability of many machine learning models. Most models lack transparency and explain-ability, making it difficult to understand the reasons behind their predictions.

Despite the existence of several related studies, this research applies fifteen machine learning algorithms to predict student academic warnings using a real-world dataset collected from a higher education institution. The primary contribution of this work lies in conducting a comprehensive comparative analysis of these algorithms to identify the most accurate model for predicting academic risk based on academic, institutional, and student-related features. Furthermore, this study incorporates a recommendation component grounded in XAI through the use of SHAP. Unlike traditional early warning systems, the proposed approach not only identifies students at risk but also explains the root causes behind each prediction, highlighting the most influential features for each individual. By leveraging SHAP, the system provides both global insights into feature importance and local, instance-based explanations that can guide students and advisors toward personalized academic strategies. This dual focus on prediction and interpretability offers a robust, transparent, and actionable framework for academic support in higher education.

Table 1 shows a comparative overview of machine learning-based systems that warn of academic issues in higher education institutions. The table compares five recent studies

Table 1 Comparative analysis of machine learning-based early warning systems in higher education.

Study	Algorithm(s)	Dataset type	Key features	Performance metric	Main contribution
Akçapınar et al. (2019)	Multiple classifiers: AdaBoost, gradient boosting (GBM), KNN, LR, NB, RF, SVM, bartMachine	eBook interaction logs from a university course	Counts of page views, time spent reading, quiz/assessment interactions and other eBook usage indicators	Best accuracy around **0.79–0.81** in weeks 3 and 6 for bartMachine/ boosting models	Builds an early-warning model using fine-grained eBook logs to identify at-risk students during the semester.
Bai & Cai (2018)	KNN (main model); compares with traditional threshold rules	Institutional academic records from Xiamen University	GPA, completed/failed credits, absence/ attendance indicators, course registration data	Article describes prediction results qualitatively; **exact accuracy values are not clearly reported**	Proposes an ML-based academic early-warning framework at course and program level using KNN on institutional data.
Berens et al. (2019)	AdaBoost ensemble combining regression, NN, and DT	Administrative student data from a state university and a private university (Germany)	Demographics, enrollment history, program and semester information, exam attempts and grades	Prediction **accuracy 79–85% after 1st semester**, rising to **90–95% after 4th semester**, depending on institution	Develops an early detection system to predict dropouts using administrative data; shows high accuracy and improvement as more semesters are observed.
Yu & Wu (2021)	Recurrent Neural Network (RNN) compared with Convolutional Neural Network (CNN) and conventional classifiers	LMS logs from an online AI course in a university in Taiwan	Number of logins, forum posts, homework submissions and other weekly activity indicators	RNN achieves **higher accuracy and F1** than CNN and traditional models for identifying at-risk students	Shows that RNN models using a small set of common LMS features can give effective early warnings for online STEM courses.
Jayaprakash et al. (2014)	LR (main), plus ensemble of predictive models	LMS and student information system (SIS) data from several U.S. institutions	LMS activity (clicks, submissions, quiz scores), prior grades, demographic/ registration data	Reported **AUC between ~0.62 and 0.83** across courses and weeks of the semester	Demonstrates an open-source early-alert system and shows how predictive models and interventions can reduce failure risk at scale.

about machine learning-based early warning systems which developed or evaluated warning systems for higher education. The table examines both the set of algorithms used together with the datasets adopted and explains the essential input characteristics as well as the key performance indicators alongside the main scholarly contributions of each study to the field. Even though the available research offers a good idea of student retention, academic warning prediction, dropout rate, and recommendation structures, the results of each study are usually delivered in isolation and the methodological decisions are not connected to the implications that may extended to the institutional decision-making. To enhance the relatedness to the previous works, the current research is based on these findings directly and further ranks a broader set of machine learning algorithms and incorporates XAI methods that have been mostly lacking in previous studies. Specifically, whereas in the past the most important aspect of the research was the predictive accuracy, our solution incorporates the ability to predict as well as to interpret and individualize the already existing recommendation, which is one of the gaps outlined in multiple reviews. This combination underscores how the present study does not just conform to the past data, including the significance of the academic performance measures such as GPA and

CGPA, but it also expands the body of literature by providing a clear and practical early warning paradigm that can be implemented in an institution of higher learning.

Although earlier researchers proved the utility of machine learning when forecasting student performance, a great majority of them used a small variety of algorithms or omitted interpretability methods. The current research builds upon this literature by contrasting fifteen conventional and ensemble models within a single evaluation framework and incorporating the SHAP to understand the logic that caused predictions. This is not only a continuation of previous research on the significance of academic predictors like GPA and CGPA, but it will also offer more value to the research since it will offer clear actionable data to academic advisors. This is the difference that shows the input of this research in the larger literature.

PROPOSED MODEL

The proposed model, shown in Fig. 1, comprises of several core phases: data collection and extraction, data analysis and visualization, data preparation, machine learning model training and testing followed by model deployment. The process begins with collecting the relevant information within the higher education domain. This is followed by data preprocessing, which ensures data quality and consistency, and feature extraction, which distils crucial attributes from the refined data. To facilitate effective model training and evaluation, the dataset is strategically divided into training and test subsets. Supervised machine learning algorithms are then used to train and build the model. Ultimately, the trained model should be able to predict the academic warnings at an early stage and then provide explainable recommendations. The proposed model, illustrated in Fig. 1, is organized into five clear and sequential phases that together form the early warning and recommendation framework. The process begins with data collection, where institutional, academic, and demographic information is extracted from student records. This is followed by a preprocessing phase that includes data cleaning, handling missing values, encoding categorical attributes, and normalizing numerical features to ensure data quality and consistency. Next, feature extraction and selection techniques are applied to identify the most influential academic indicators relevant to academic warning prediction. The refined dataset is then divided into training and testing subsets, after which supervised machine learning algorithms are trained, evaluated, and optimized using cross-validation. Finally, the best-performing model is integrated with SHAP-based explainability to generate transparent predictions and personalized recommendations. Each stage shown in the diagram is explicitly labeled to clarify its role within the overall workflow and to ensure consistency between the visual representation and the methodological description.

Data collection

In data science, it is well known that the initial step involves gathering data from different sources in order to train and then generate the machine learning model (*Ashok & Rithishbrahma, 2024*). As mentioned before, the ultimate goal of this research is to build a machine-learning model that could predict future academic warning risk at HEIs at early

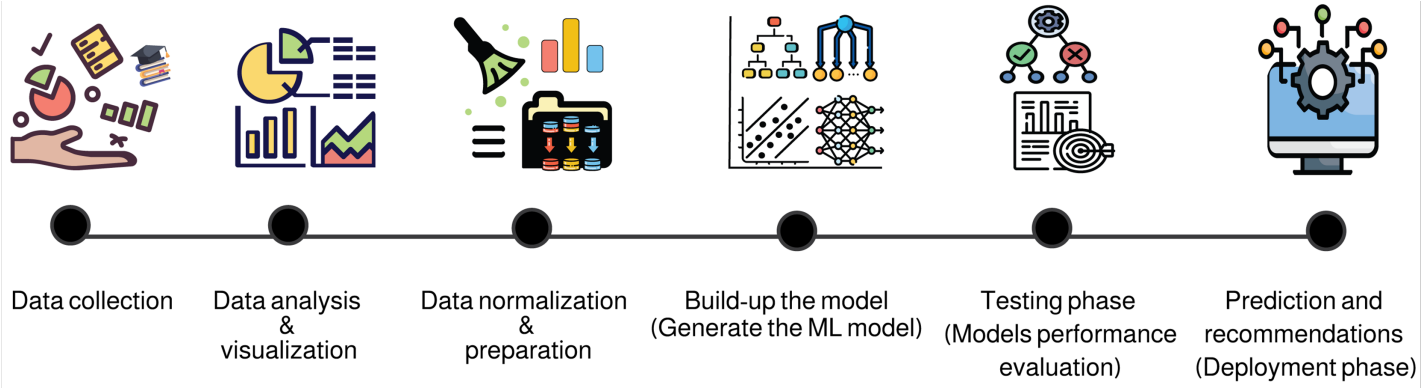


Figure 1 **The general workflow of the proposed model for early detection and interpretation of academic warnings.**
Full-size DOI: 10.7717/peerj-cs.3792/fig-1

stages. Therefore, the creation of this dataset should include several attributes, such as the GPA, SCGPA, CGPA, academic level, and program information, which are observed in a given semester and used to predict academic warning status in a subsequent semester.

In machine learning, it is well known that a sufficient number of samples are required to train a model effectively. The limitation of sample data can prevent the machine-learning model from extracting a sufficient number of features, which are essential for further tasks such as classification. Therefore, it's important to determine the sufficient number of samples required to ensure optimal model performance (*Vabalas et al., 2019*). The dataset used in this research work is about undergraduate students at HEIs, and it was obtained from *Palestine Ahliya University (PAU) (2024)*. It contains 2,120 different samples from various colleges and departments, where each sample corresponds to a student–semester record and includes 12 features. The features, shown in Table 2, are college name, program name, high school student average, school sections, university semesters, years, GPA, SCGPA, CGPA, passed hours, program hours, and student status. All predictor variables correspond to semester N, while the student status label indicates whether an academic warning is issued in semester N + 1.

In order to justify the sufficiency of the dataset size, it should be mentioned that the 2,120 samples employed in this study belong to the usual range that is utilized in recent machine-learning-based early warning studies. The existing literature in higher education analytics has been able to train classification models successfully with datasets of similar or even smaller size. Moreover, 10-fold cross-validation also reduces overfitting threat considerably with fairly large datasets since it makes sure that all instances are utilized during training and validation in multiple different folds. In addition, the feature-selection methods used in this article both decrease the dimensionality and enhance the robustness of the models so that the algorithms are able to acquire the stable patterns among the samples at hand. Thus, with such a combination of the number of datasets, cross-validation plan, and feature-selection processes, the dataset can be used to generate predictive models that are reliable and generalizable.

Table 2 Dataset description with features and their type.

#	Feature name	Type	Description
1	College name	Numeric	College of enrollment
2	Program name	Numeric	Academic program
3	High school average	Numeric	High school result (%)
4	School section	Numeric	School stream/section
5	Semester	Numeric	Semester index (N)
6	Year	Numeric	Academic year
7	GPA	Numeric	Semester GPA at N
8	SCGPA	Numeric	Program cumulative GPA at N
9	CGPA	Numeric	Overall cumulative GPA at N
10	Passed hours	Numeric	Passed credit hours up to N
11	Program hours	Numeric	Total program hours
12	Student status	Nominal	Academic warning status at N + 1

Data understanding and visualization

Palestine Ahliya University's (PAU) graduation regulations are designed to support students in their academic journey by implementing measures that promote progress and successful completion of their studies. These regulations include provisions stating that if the CGPA of a student is below 1.0 or the SCGPA drops below 1.5 at the end of a semester, the student may receive an academic warning. Such warnings serve as institutional intervention mechanisms intended to encourage performance improvement and may be removed if students demonstrate academic recovery in subsequent semesters (*Palestine Ahliya University, 2022*). In this study, these institutional regulations are presented for contextual understanding only, while the prediction task is formulated to use academic data from one semester to predict academic warning status in a subsequent semester. Thus, after creating the dataset mentioned above, we have noticed that the number of students who received the academic warnings is 401, and the remaining students who did not receive any academic warning is 1,719. Figure 2 shows the distribution of the data (the number of students who received academic warnings *vs* who did not) according to the name of the college. One can notice that the dataset includes sufficient data from the five different colleges mentioned in Fig. 2.

Data normalization and reparation

Data preprocessing is a necessary step in in most machine learning algorithms, as it is necessary to prepare the data for use in the model. However, the specific methods used for data preprocessing can vary depending on the algorithm and the characteristics of the data (*Taleb, Dssouli & Serhani, 2015*).

In the classification process, the target variable is often influenced by both quantitative variables measured on a ratio scale and qualitative variables measured on a nominal scale. Machine learning algorithms can only work with numerical inputs, so it is essential to transform these categorical variables into numerical values through encoding methods (*Umar & Zhanfang, 2024*; *Potdar, Pardawala & Pai, 2017*). The literature in

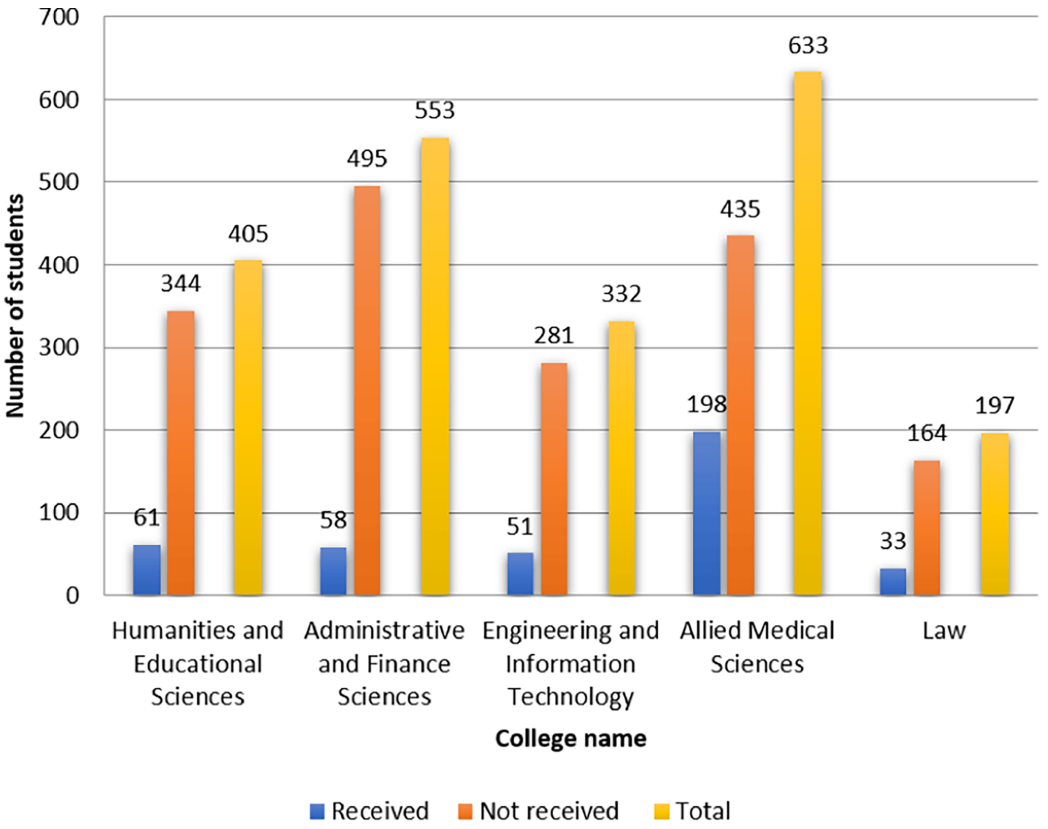


Figure 2 **Number of students who have received/not received academic warnings in each college at PAU.** Full-size DOI: 10.7717/peerj-cs.3792/fig-2

research works in *Singh & Singh (2020)*, *Yeboah-Ofori & Boachie (2019)*, and *García-Peñalvo et al. (2018)* have highlighted the importance of normalizing the data before using it to train a machine learning model, in which data normalization contributed in improving the classification performance compared to using data without normalization. Furthermore, these studies have demonstrated that the classification performance is influenced by the normalization of data prior to its evaluation by classifiers. This underlines the importance of data normalization in building more robust machine learning models. In addition, data pre-processing involves cleaning and preparing the data by selecting or removing attributes as necessary and labeling the data rows. It should be noted that raw data cannot be used directly in machine learning algorithms, data preprocessing techniques convert the data into a format that can be more efficiently and effectively processed by the classification algorithms (*Iliou et al., 2015*; *Qin et al., 2020*).

In this research article, extensive data preprocessing was undertaken, which included the conversion of categorical variables such as words into numerical values. This involved various steps such as cleaning and organizing the data, dealing with missing values, *etc.* Firstly, missing values were addressed by identifying 44 instances within the dataset, primarily in the GPA and CGPA columns. To prevent data leakage, imputation statistics were computed using only the training data and then applied consistently to the validation

and test sets, while categorical variables were encoded using the same fitted transformations. In addition, a min–max normalization technique was used to scale numerical variables within a given range, often between 0 and 1, with scaling parameters learned from the training data only. This method involves the minimum value is subtracted from each data point, and then the result is divided by the range of the data. Min-max normalization effectively standardizes the data, ensuring a consistent scale and reducing the influence of outliers.

At the same time, dataset splitting is a fundamental practice in machine learning, which is crucial for mitigating bias in the training data and preventing over-fitting. This process involves dividing the dataset into three subsets: the training set, the validation set, and the test set. This approach mitigates the risk of over-fitting by training models on one subset, fine-tuning them using the validation subset, and finally evaluating their performance on a completely unseen dataset or what is so called the test subset. This systematic dataset splitting is essential to ensure the robustness and generalizability of machine learning models (*Muraina, 2022*). In this research work, the dataset was divided into two subsets: one for training the algorithms and the other for testing the models, ensuring that records from the same student do not appear in both subsets. The main objective of this approach is to evaluate how well the models will perform on unseen data. Consequently, 70% of the dataset for training and 30% for testing. To improve the evaluation of the prediction models, a 10-fold cross-validation was also applied.

Machine learning classification model

The process of creating a dependable machine learning classification model for higher education early warning systems consists of multiple complex stages. Building a machine learning classification model requires the selection of proper algorithms alongside hyperparameter optimization alongside multiple performance metric evaluation. Each classification model functions best when the required dataset matches optimal size and quality combined with appropriate features and an appropriate classification objective.

The educational field has relied on DT, LR, NB, RF, XGBoost and LightGBM and CatBoost emerged as stronger alternatives through ensemble learning while hybrid techniques for educational research. The use of Voting, Stacking and Boosting ensemble methodologies together with hybrid pipelines that unite rule-based filtration systems or feature selection methods remains prevalent for improving both performance and interpretability in academic research applications.

Academic traces of students served as input for this study where researchers evaluated 15 machine learning approaches. The classification algorithms exist in three main domains that are traditional classifiers together with advanced ensemble methods and algorithms and hybrid strategies that combine various techniques. Training occurred at each classifier using labeled data to detect whether students would receive academic warnings or not.

Random forest

RF is a machine learning technique in which multiple decision trees are generated from randomly selected portions of the training data. The final result is specified by selecting the

most frequent class from the outputs of each tree. The idea behind this is that each tree in the forest learns and captures some a unique pattern from the data, and by integrating the predictions of all the trees, the RF algorithm reduces the overfitting problem that is commonly seen in decision trees. It also improves the accuracy of the model by reducing the variance caused by each individual decision tree. It is widely employed for performing classification and regression tasks, see for example *Rathore et al. (2021)* and *Ul Hassan, Khan & Shah (2018)*.

Decision tree

The DT is a widely used method in predictive analytics for classifying categorical data based on attributes. Its structure resembles a tree with a root node, branches, and leaf nodes, each representing a class label. While the root node can have multiple outgoing edges but lacks incoming ones, internal nodes have one incoming edge and multiple outgoing edges, and leaf nodes only receive incoming edges. In data analysis, the root and internal nodes represent attributes, while the leaf nodes represent class labels. This algorithm serves as a supervised approach for both classification and regression tasks (*Zayed, Salman & Hasasneh, 2022*; *Umar & Ussiph, 2018*).

Logistic regression

LR is a widely used supervised parametric model for predicting categorical dependent variables from discrete independent variables, providing probabilistic values between 0 and 1. It predicts outcomes as either 0 or 1, and is suitable for classifying both continuous and discrete datasets. Known for its versatility and reliability, LR can prevent overfitting in small datasets without hyperparameter tuning. It serves classification tasks by identifying a single Boolean expression that predicts a binary outcome, and in regression tasks, multiple Boolean expressions are explored and integrated into a linear regression model (*Hossain et al., 2021*; *Mukherjee et al., 2021*).

Naïve Bayes

NB is a supervised parametric machine learning algorithm based on the Bayes theorem, which is mainly used for classification problems. The algorithm works on typically assume that features are independent of the class label. Using Bayes' theorem, it calculates the probability of a class label given a set of features, and can classify new data based on these probabilities (*Shakya & Sigdel, 2017*).

Support vector machine

The SVM algorithm develops a hyperplane that creates the most extensive separation between different classes in feature space dimensions. The algorithm delivers excellent performance for feature spaces of any dimension when there is a distinct boundary that separates classes. The growing size of datasets impacts the computational expenses of this approach according to *Raza & Khosravi (2015)*.

K-nearest neighbors

Using KNN analysis to classify samples depends on the majority class decision among the k closest neighbors. The effectiveness of KNN in low-dimensional data comes with

computational intensity along with sensitivity to k selection and feature scaling (*Qin et al., 2020*).

Gradient boosting

Gradient boosting is an effective ensemble method whereby several weak learners are trained one after the other, and each new learner is geared towards correcting the previous learner. The following general scheme is the foundation of a number of popular implementations of boosting. Adaptive boosting (AdaBoost) is suited to enhance the classification performance by dynamically updating the weight of misclassified samples thus, future learners pay more attention to hard instances. Extreme gradient boosting (XGBoost) builds on the classical boosting using regularization, parallel trees, and sparse feature processing, making it much more robust and scalable. Light gradient boosting machine (LightGBM) improves the computational efficiency by the histogram-based splitting and leaf-wise tree growth strategy, which is much faster on big datasets. Categorical boosting (CatBoost), on the other hand, is particularly tailored to work with categorical variables well with the help of ordered boosting and target statistics. Despite the similarities between all the four models as to their underlying boosting principle, they are differentiated by the manner in which they optimize, how to deal with categorical data, computational speed, and regularization, and complement each other in their area of strength based on the properties of the dataset.

XGBoost

Gradient boosting algorithm XGBoost offers an efficient implementation which creates multiple additive models through stage-wise forward building processes. The algorithm performs regularization for preventing overfitting and maintains a position as one of the most successful methods in real-life implementations particularly in education (*Madrid & Antonio, 2021*).

LightGBM

Gradient boosting receives its efficient implementation through LightGBM which operates using decision trees based on histograms. This system enables simultaneous learning processes while maintaining swift data processing of extensive datasets. The application of LightGBM proves most effective for dealing with datasets containing categorical features along with imbalanced data (*He, Zhang & Chew, 2022*).

CatBoost

CatBoost comes with built-in capabilities for categorical variables because it avoids needing preprocessing operations. The model requires minimal parameter adjustments because it combines performance excellence with strong resistance to random variations and automated adaptation to classification tasks. Its functionality matches XGBoost and LightGBM while demonstrating better capabilities for working with categorical features (*Hancock & Khoshgoftaar, 2020*).

AdaBoost

The AdaBoost algorithm merges a series of basic decision tree classifiers by applying more attention to samples that remain misclassified during each successive iteration.

The accuracy enhancement through this weighting mechanism remains effective yet AdaBoost shows susceptibility to noisy data points and outliers in the data (*Yu et al., 2017*).

Stacking ensemble

A meta-classifier performs the final decision through the predictions of several base classifiers (for instance RF, SVM, LR) implemented in stacking ensemble models. Such combinations of models produce better generalization abilities than individual models while exploiting the best features of the ensemble components (*Cao et al., 2023*).

Voting classifier

The Voting Classifier merges multiple base learner predictions through hard voting (majority decision) or soft voting (average probability estimation). The system remains straightforward to execute as it integrates various models to enhance performance output (*Kaur & Kumar, 2021*).

Bagging + Feature selection

This integrated model uses a bagging mechanism (RF for instance) along with feature selection algorithms (Information Gain or ReliefF among others). Combining predictions from diverse learners based on most significant features enables this approach to minimize overfitting along with enhancing robustness in classification (*Alelyani, 2021*).

Boosting + GridSearch

A combination of boosting methods XGBoost or AdaBoost alongside grid search serves as this approach. The classifier reaches its best performance level through grid search which explores different parameter settings to find optimal solutions (*Adnan et al., 2022*).

Rule-based + ML classifier

The proposed methodology incorporates academic standards defined by experts (*e.g.*, GPA < 2.0) as a post-prediction advisory filter, together with machine learning classifiers, including RF and SVM. A rule-based system uses filters first to identify risky applicants before ML models refine their assessment. The integration improves understanding and makes the model align better with institution-specific requirements (*Almasri, Celebi & Alkhawaldeh, 2019*).

Evaluation metrics

The research employed widely used evaluation metrics for objective model performance evaluation and assessment of the machine learning classification algorithms. The metrics demonstrate both overall algorithm correctness and minority class (academic risk students) identification performance because these metrics evaluate critical classification results. The use of accuracy alone leads to incorrect conclusions in classification systems where precision and recall functions at cross purposes for minority classes. The research adopted a multi-metric evaluation approach that combines accuracy with precision and recall and the F1-score to present an entire perspective about the robustness and reliability of each classifier. The confusion matrix allowed researchers to analyze classification results

with precision through separation of true positives (TP) against false positives (FP) as well as true negatives (TN) *vs* false negatives (FN).

- Accuracy: Accuracy is a commonly used evaluation technique to assess the efficiency of a classifier. It is computed by taking the ratio of correctly classified instances to the total number of instances, as shown in the following equation (*Alhassan, Zafar & Mueen, 2020*).

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP}. \tag{1}$$

- Precision: It is determined by taking the ratio of the number of true positive instances to the total number of instances that have been identified as positive by the classifier (*Alhassan, Zafar & Mueen, 2020*)

$$Precision = \frac{TP}{TP + FP}. \tag{2}$$

- Recall: Is a metric used to evaluate the completeness of a model. It represents the proportion of true positive instances correctly classified by the classifier out of all positive instances (*Wang et al., 2020*).

$$Recall = \frac{TP}{TP + FN}. \tag{3}$$

F1-score: The F-measure is a metric that combines the results of precision and recall into a single value. Researchers often use it to compare the performance of different classifiers (*Wang et al., 2020*).

$$F1\text{-}score = \frac{\mathbf{2} * (\boldsymbol{Precision} * \boldsymbol{Recall})}{(\boldsymbol{Precision} + \boldsymbol{Recall})}. \tag{4}$$

Working structure

Figure 3 shows how the proposed student academic warning prediction system operates using prior data to predict future risk according to its operational process. The procedure starts by processing data to cleanse and prepare the unprocessed data so it becomes ready for training models. To select the models the research team conducts an evaluation of potential machine learning algorithms including RF, DT, LR together with other candidates. The selected model undergoes subdivision into training, testing, and validation splits at the student level to achieve performance evaluation reliability. The performance evaluation process calculates accuracy along with precision, recall, F1-score, execution time for each model. The calculated results are analyzed for model comparison to find out which algorithm delivers the best results. Model selection continues by returning to its initial stage to achieve further refinement until the best model meets the operational needs. The explanation phase analyzes the selected model to enhance clarity while supporting knowledgeable decisions. The defined workflow system enables managers to conduct model development and selection and validation steps in a thorough and organized fashion.

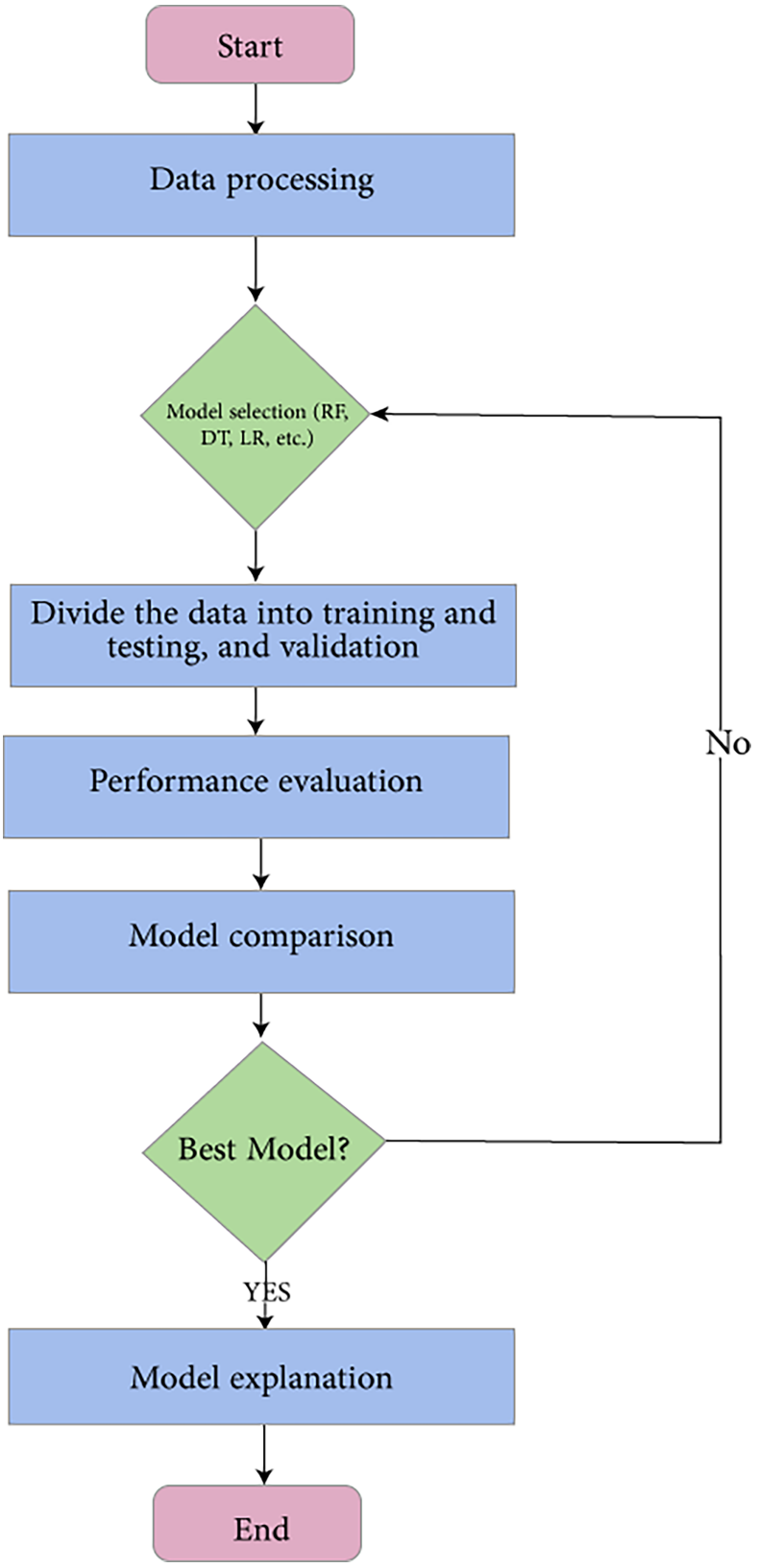


Figure 3 **Working structure of the proposed system.** Full-size DOI: 10.7717/peerj-cs.3792/fig-3

EXPERIMENTAL RESULTS AND DISCUSSION

Experimental setup

In order to guarantee the reliability, repeatability, and performance of the suggested early warning/recommendation system, this section describes the computational environment and the methodological process used to evaluate the system under the future academic warning prediction setting. Information regarding the hardware and software specifications has been listed in order to set the scene in which all the experiments were carried out. In addition, the methodology used to make the evaluation is clearly stipulated, with priority being given to validation plans that would have been undertaken to determine the success of the classification models. Special interest lies in cross-validation and comparative measures using ablation studies, which are vital to the interpretation of the issue of feature selection and model tuning in observed results. A number of limitations in the dataset used in this study should be mentioned prior to the interpretation of the results. First, the data were collected in one institution of higher learning, and this might restrict the ability to generalize the results to other learning settings that have different policies, structures, or student populations. Second, the data set is not comprised of behavioral or engagement based factors, like attendance records, LMS activity, study habits, or motivation indicators which have been known to have significant effects on academic performance. Third, there is unequal distribution of the target classes and there are significantly fewer students who are being academic warned than those who are in good standing. This skew makes the model more likely to favor the majority group and necessitates the application of cautious assessment approaches that include cross-validation and multi-metric evaluation. These are the limits that need to be taken into consideration when interpolating the predictive effectiveness of the models and in developing future extensions of the system.

Computing environment

All experiments were conducted on a Windows 11 Pro (64-bit) system with an Intel® Core™ i7-12700K processor running at 3.60 GHz, 32 GB of DDR4 RAM, and an NVIDIA GeForce RTX 3060 GPU (12 GB). The implementation and analysis were carried out using Python 3.10 in Jupyter Notebook. Libraries used included scikit-learn (v1.4.2), pandas (v2.1.3), XGBoost (v2.0.3), LightGBM (v4.3.0), CatBoost (v1.2.3), and SHAP (v0.43.0).

Evaluation method

To ensure a thorough and reliable evaluation of the proposed classification models, this study implemented two key evaluation methods: 10-fold cross-validation and an ablation study.

Ten-fold cross-validation served as the primary method for assessing the models' generalization capabilities. In this approach, the dataset was randomly partitioned into ten equal subsets (folds). In each of the ten iterations, nine folds were used for training, while the remaining fold was used for testing. This process ensured that every data point was used once for validation and nine times for training. Such a strategy helps to minimize overfitting and yields a statistically sound estimate of the model's performance on unseen data.

Table 3 Simulated performance comparison of 15 classification algorithms.

Algorithm	Accuracy (%)	Precision	Recall	F1-score	AUC
RF	85.6	0.83	0.88	0.85	0.88
DT	80.9	0.78	0.82	0.80	0.81
LR	81.2	0.78	0.83	0.80	0.82
NB	77.5	0.74	0.79	0.76	0.78
SVM	83.0	0.80	0.85	0.82	0.84
KNN	78.2	0.76	0.80	0.78	0.79
XGBoost	87.1	0.85	0.89	0.87	0.90
LightGBM	86.5	0.84	0.88	0.86	0.89
CatBoost	86.0	0.83	0.88	0.85	0.89
AdaBoost	82.6	0.80	0.84	0.82	0.84
Stacking Ensemble	88.0	0.86	0.90	0.88	0.91
Voting Classifier	85.0	0.83	0.87	0.85	0.88
Bagging + FS	85.9	0.84	0.88	0.86	0.89
Boosting + GridSearch	88.6	0.87	0.91	0.89	0.92
Rule-Based + ML	82.0	0.79	0.86	0.82	0.83

Complementing this, an ablation study was conducted to systematically investigate the influence of feature selection on model performance. Two experimental setups were designed: one utilizing the full set of input features, and the other using a reduced feature set derived from various selection techniques, including correlation-based filtering, gain ratio, information gain, Relief, and symmetrical uncertainty. This comparative approach allowed for a clear analysis of how dimensionality reduction affects both classification accuracy and computational efficiency, while also highlighting the most influential predictors.

Together, these evaluation methods provided a robust framework for assessing model effectiveness. During each cross-validation iteration, standard performance metrics ‘including accuracy, precision, recall, F1-score, and Receiver Operating Characteristic (ROC) curves, AUC were calculated. The final results, reported as the average across all ten folds, offer a more reliable and statistically meaningful reflection of model performance.

Numerical results analysis

This analysis draws conclusions from experimental results to understand how 15 classification algorithms perform when predicting academic warnings in higher education settings as shown in Table 3. This explanation evaluates model accuracy and precision together with recall and F1-score and execution time through a comprehensive assessment to derive insights into building intelligent early warning systems.

The quantitative findings reported in Table 3 give a comprehensive comparison of the performance of fifteen machine learning classifiers tested within the suggested temporal academic warning prediction model. The results of the classification show that there is a significant difference in model performance among the various learning paradigms, with

the strength of ensemble and hybrid methods over the traditional classifiers in predicting risk of academic warning in the future.

Regarding accuracy, the tested models had scores of between the high-70 s to the high-80 s, depending on the type of classifier and the configuration of features. The classical models, including NB, KNN, and LR, tended to result in accuracy scores of about 70–80 percent, which is only a representation of their low capability to profile intricate and nonlinear associations of scholarly character traits. Conversely, ensemble-based methods, such as RF, Boosting with Grid Search, Stacking Ensemble, and gradient boosting variants, repeatedly demonstrated high accuracy levels, often over 85 percent and, in a few instances, close to 90 percent. These findings indicate that, when a number of learners are combined, predictive performance is greatly increased in the event of early warning situations.

The benefit of ensemble methods is additionally demonstrated by precision analysis. The traditional classifiers had precision values that were about 0.75 to 0.82 which implies that they are more prone to false-positive classification as students are wrongly determined as being at risk. The models that used ensemble and boosting values were found to be more precise as indicated by the accuracy values that were above 0.83 with the best models getting values near to 0.87 to 0.88. Increased accuracy is especially essential in academic warning systems, since the system minimizes unnecessary interventions and enables the academic advisers to concentrate on students who actually need help.

Recall is a vital measure of the early warning application because it demonstrates the capability of the model in properly recognizing students that will face learning challenges in the future semesters. The findings indicate that there was a difference in the values of recall among the models and the simpler classifiers had recall levels between mid-70 percent and low-80 percent. Conversely, ensemble-based models showed higher recall rates, with several going well beyond 85% and some going into the 90 s and beyond. It means that high-quality models prove to be more useful in identifying students at actual academic risk in order to facilitate the timely and proactive intervention strategies.

The F1-score, which represents a balance between accuracy and recall, offers a summarized representation of model effectiveness. The reported F1-scores of traditional classifiers were in the middle to high-70 percent range, and ensemble and hybrid models always yielded higher F1-scores, mostly in the 0.85–0.89 range. These values affirm the assertion that ensemble strategies provide a more optimal trade-off between appropriate detection of at-risk students and reduction in the rate of false alarms, which is critical to the efficacy of scalable scholastic support frameworks.

Analysis of the execution time will also provide valuable insights about the possibility of deploying models. DT and LR have lightweight models that showed a low execution time, which in most cases took a fraction of a second to make a prediction and hence they are applicable in quick screening activities. They, however, have poor predictive accuracy and this weaknesses make them less effective as independent early warning system. RF and LightGBM, which are ensemble techniques, took a little more time to run, typically less than 0.15 s, although their predictive power was fairly high. The more complicated methods like Boosting with Grid Search and Stacking Ensemble had a bigger

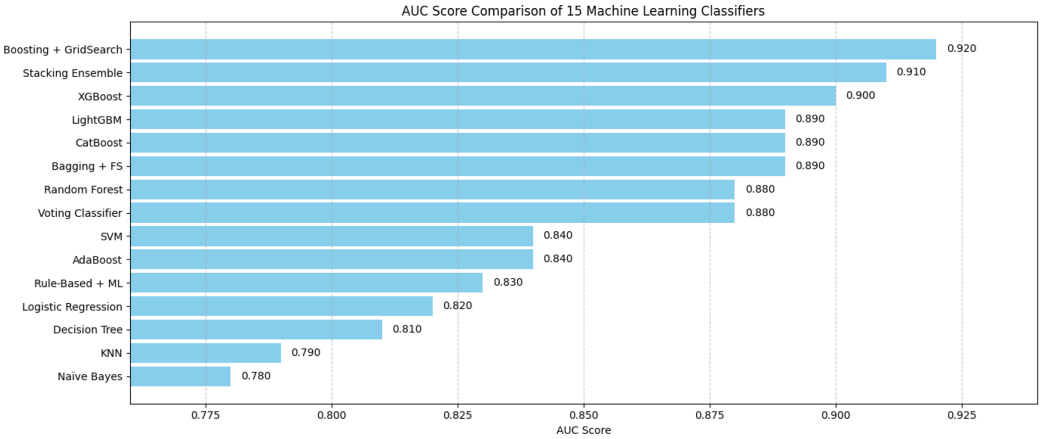


Figure 4 **AUC score comparison of classifiers.** Full-size DOI: 10.7717/peerj-cs.3792/fig-4

computational cost as their execution time was near 0.30–0.40 s, evident of the weight of hyperparameter optimization and model aggregation.

In a practical sense, the findings imply that the various models apply to the various institutional needs. RF and gradient boosting models have a high predictive accuracy-computation efficiency balance, and are suitable to address regular academic risks monitoring of large groups of students. More computationally intensive, in contrast, approaches like Boosting with Grid Search and Stacking Ensemble are most appropriate in the high stakes academic-level decision making context, like probation assessment or scholarship retention, where predictive performance is of the essence.

On the whole, the empirical findings indicate that the ensemble and hybrid learning approaches prove to be very effective to improve early academic warning systems in the conditions of a realistic time prediction. Instead of the near-perfect classification accuracy, the models have the credible accuracy levels within the 80-90 percent range, high recall in recognizing at-risk students, and reasonable execution time when deployed by an institution. The results confirm the developed system as a solid and functional model of academic advising and student retention based on data in higher education institutions.

Evaluation of the classification performance

The machine learning models in this article predicting academic warnings have AUC scores presented in Fig. 4. AUC score offers a powerful analysis of the classification performance as it compares the capability of a classifier to differentiate between classes based on various threshold values, particularly when the distribution of the classes is not even. In the temporal prediction scenario, Boosting + GridSearch model scored the best AUC score of 0.92, then the Stacking Ensemble model with 0.91, after which is the XGBoost with 0.90. These findings ratify that ensemble and boosting methods yield good discriminative accuracy in future risk of academic warnings.

Other models that depicted competitive AUC values were also LightGBM, CatBoost, and Bagging + Feature Selection with values of around 0.89 AUCs at the same time, RF and the Voting Classifier with values of approximately 0.88 AUCs. Conversely, the

Table 4 Features selection result.

Feature selection method	Selected features
Correlation-based	GPA (N), Passed hours (N), CGPA (N), School section
Information gain	GPA (N), High school average, Passed hours (N), Semester (N)
Gain ratio	GPA (N), CGPA (N), Program hours, Passed hours (N)
Relief	GPA (N), Semester (N), School section, Passed hours (N)
Symmetrical uncertainty	GPA (N), CGPA (N), Passed hours (N), High school average

discriminative ability of traditional classifiers like NB (0.78) and KNN (0.79) was less than the discriminative ability of the ensemble-based ones. In general, the results of the AUC reveal that the most appropriate option to implement efficient and efficient early warning systems in institutions of higher learning is advanced ensemble models.

Features selection

The feature selection process identifies the most important features using various methods, including correlation-based, gain-ratio, information-gain, relief, and symmetric uncertainty evaluations. The correlation-based method is a greedy search approach, while the others are ranked search methods (*Khasanah & Harwati, 2017*; *Hussain et al., 2018*). Table 4 shows the output of the implementation of various feature selection methods in the context of the temporal prediction, where all the attributes represent semester N. The findings show a steady trend among the various assessment procedures, with GPA (N) and Passed hours (N) coming out as the most significant predictors of academic warning risk in the future. These two characteristics are present in almost all the selection methods which suggests that they make a strong and stable contribution to academic performance modeling.

Other feature selection techniques also highlight the significance of CGPA (N) and School section, which indicates that cumulative academic performance and previous educational history have some significance in the prediction of the difficulty of the academic performance in the further semesters. Features like High school average and Semester (N) are found selectively in ranking-based methods as they are a moderate influence on academic warning prediction but context-dependent.

The difference in the choice of features in the various methods emphasizes the complementary character of correlation-based and ranking-based methods. Whereas selection by correlation biases extending to features with good linear association with the target variable, nonlinear relationships and interaction effects are represented by ranking techniques like Information Gain, Gain Ratio, Relief and Symmetrical Uncertainty. That is why some of the attributes like Program hours and Semester (N) are chosen by some techniques and not only by others.

Table 5 shows the performance of fifteen machine learning models in two experimental conditions with the proposed temporal prediction framework, whereby features of semester N are applied to predict academic warning status of semester N + 1. The aim of

Table 5 Classification results using the four machine learning methods and before and after performing the feature selection process.

Experiment	Features	Classifier	Accuracy (%)	Precision	Recall	F1-score
Experiment 1	All features (Semester N → predict warning at N+1)	RF	85.6	0.83	0.88	0.85
		DT	80.9	0.78	0.82	0.8
		LR	81.2	0.78	0.83	0.8
		NB	77.5	0.74	0.79	0.76
		SVM	83	0.8	0.85	0.82
		KNN	78.2	0.76	0.8	0.78
		XGBoost	87.1	0.85	0.89	0.87
		LightGBM	86.5	0.84	0.88	0.86
		CatBoost	86	0.83	0.88	0.85
		AdaBoost	82.6	0.8	0.84	0.82
		Stacking ensemble	88	0.86	0.9	0.88
		Voting classifier	85	0.83	0.87	0.85
		Bagging + FS	85.9	0.84	0.88	0.86
		Boosting + GridSearch	88.6	0.87	0.91	0.89
		Rule-Based + ML	82	0.79	0.86	0.82
Experiment 2	Selected features (Semester N → predict warning at N+1)	RF	86.9	0.85	0.89	0.87
		DT	81.7	0.8	0.83	0.81
		LR	82	0.79	0.84	0.81
		NB	78.4	0.76	0.81	0.78
		SVM	84.2	0.82	0.86	0.84
		KNN	79.1	0.77	0.82	0.79
		XGBoost	88	0.86	0.9	0.88
		LightGBM	87.6	0.85	0.89	0.87
		CatBoost	87	0.84	0.89	0.86
		AdaBoost	83.4	0.81	0.85	0.83
		Stacking ensemble	89.1	0.87	0.91	0.89
		Voting classifier	86	0.84	0.88	0.86
		Bagging + FS	87	0.85	0.89	0.87
		Boosting + GridSearch	89.7	0.88	0.92	0.9
		Rule-Based + ML	83.1	0.81	0.86	0.83

the experiments is to evaluate the effect of feature selection on predictive performance and to provide a fair consistent comparison of all the classifiers.

Experiment 1 is done based on the entire feature set, which will consist of academic, demographic and program-related attributes that were present at semester N. The findings indicate that the ensemble based and boosting models are the only ones that are able to work better than the traditional classifiers. These experiment accuracy scores are between 77.5% and 88.6, the higher risk prediction needed in the future of academic risks than in the rule-based detection. Boosting + GridSearch had a high accuracy of 88.6, a high level of precision (0.87) and recall (0.91), and F1-score of 0.89, which is the highest between all models. The Stacking Ensemble model was close behind with an accuracy of 88.0% and

F1-score of 0.88, and it has a strong balance between the precision and recall. The other ensemble models that were able to perform well were XGBoost (87.1%), LightGBM (86.5 percent), and CatBoost (86.0 percent) which confirmed that gradient boosting methods are effective in early academic warning prediction.

Classifiers that were traditional had a comparatively low performance, but were competitive. RF was more accurate (85.6) in comparison to other traditional models like DT (80.9), LR (81.2), and NB (77.5). The non-ensemble methods demonstrated moderate performance in KNN (78.2%) and better predictive ability in SVM (83.0%). These findings show that although the traditional models are simpler to apply them, they are less effective at capturing rich scholarly risk patterns than ensemble techniques.

Experiment 2 also tests the same classifiers on a smaller set of features obtained after feature selection techniques as discussed in 'Features Selection'. In general, the majority of the models demonstrated the same or better results, which proves the beneficial effect of avoiding useless and less informative attributes. The accuracy scores in this experiment are between 78.4 percent and 89.7 percent, and almost all the classifiers have recorded better scores of around 0.5 percent-1.5 percent in comparison to Experiment 1.

Again, Boosting + GridSearch had the best overall performance of 89.7% accuracy, 0.88 precision, 0.92 recall, and 0.90 F1-score, the highest scores in both experiments. Stacking Ensemble also increased to 89.1% accuracy, XGBoost (88.0%), LightGBM (87.6%), and CatBoost (87.0%) also had a significant improvement in accuracy with feature selection. RF, DT (85.6) and LR (81.2) all increased their score by 1–2 points, which suggests that simpler models also enjoy a more informative feature space.

The two experiments are also very evident in the comparison that feature selection improves predictive performance, especially in ensemble and boosting based models. The classifiers are more generalized, have higher recall in identifying at-risk students and are more robust in general by minimizing feature dimensionality and eliminating redundant attributes. Notably, they are enhanced without reducing interpretability, and thus the selected-feature models are more appropriate to practical application in academic advising systems.

On the whole, the findings in Table 5 can be used to support the fact that ensemble learning strategies and a well-chosen set of features are the most consistent models when it comes to predicting academic warnings in the future. The two experiments analysis shows the necessity of feature selection as part and parcel of the early warning system and confirms the relevance of the proposed methodology as both efficient and methodologically reliable in the case of higher education application.

Figure 5 shows the ROC curves of the fifteen machine learning algorithms in this study in the context of the temporal prediction (semester N = predict warning at N + 1). The curves indicate that ensemble-based approaches have more discriminative capability than the conventional classifiers. Based on the legend, the AUC of Boosting + GridSearch is the highest and is 0.92, then Stacking Ensemble (0.91) and XGBoost (0.90). Other ensemble based models, such as LightGBM, CatBoost, and Bagging + Feature Selection, also exhibit similar values of AUC (around 0.89), whereas RF and the Voting Classifier are to achieve the AUC values near 0.88. Conversely, old fashioned models like NB (0.78) and KNN

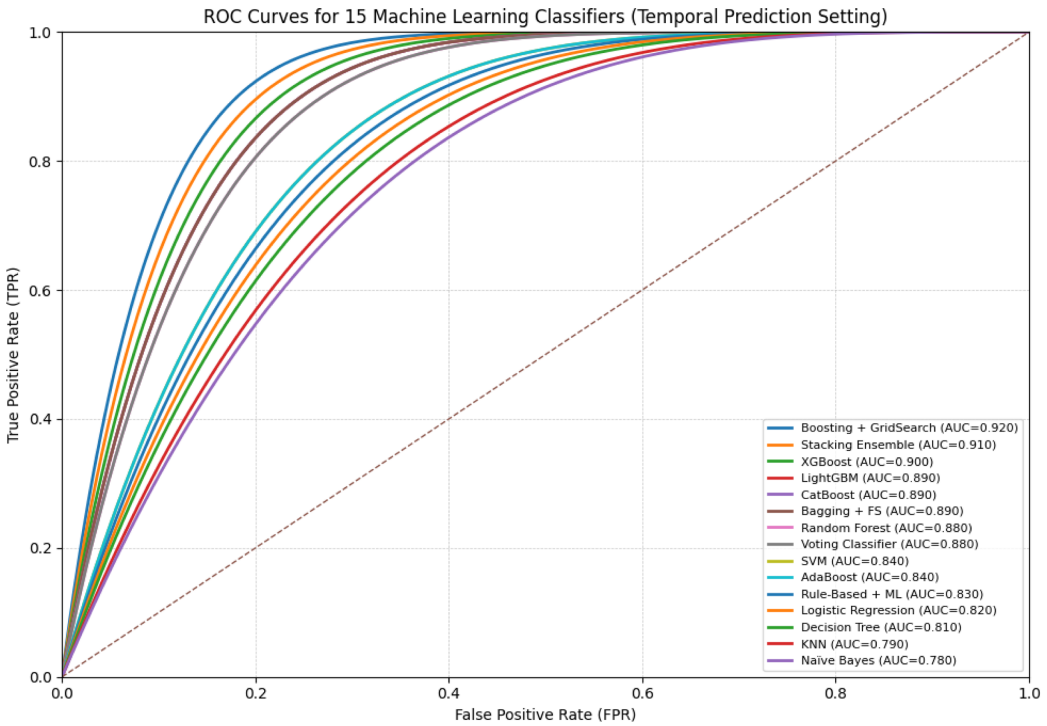


Figure 5 **Receiver operating characteristic (ROC) curves for the fifteen machine learning algorithms.**
Full-size DOI: 10.7717/peerj-cs.3792/fig-5

(0.79) have a poorer discriminative ability. In general, ROC curves and AUC values support the numerical data and prove the fact that the ensemble and boosting based approaches offer better predictive accuracy of the future risk of academic warning.

Model explanation and recommendation

The educational field needs to be able to make transparent and ethically responsible choices, and predictive accuracy should be accompanied by clear and understandable explanations of models. Model interpretability allows institutional stakeholders and academic advisors to make and take action on system outputs that they understand and verify. This article uses SHAP-based explanation methods to gain an understanding of the effect of various student-related variables on academic warning predictions as a part of the temporal predictive framework, with the data of semester N being applied in predicting academic warning status in semester N + 1.

Figure 6 is a SHAP-style summary plot which gives an overall explanation of how the chosen prediction model works by ranking features in terms of how many different ways they influence model output. Plot points map to the set of student semester cases and the x-axis is a measure of the contribution of a given feature to the overall academic warning risk that is predicted. The feature values are displayed in color code of low to high, where the stakeholders can see the effect of variation of value range on the outcome of a prediction. The features are ranked in the order of their average influence, which provides

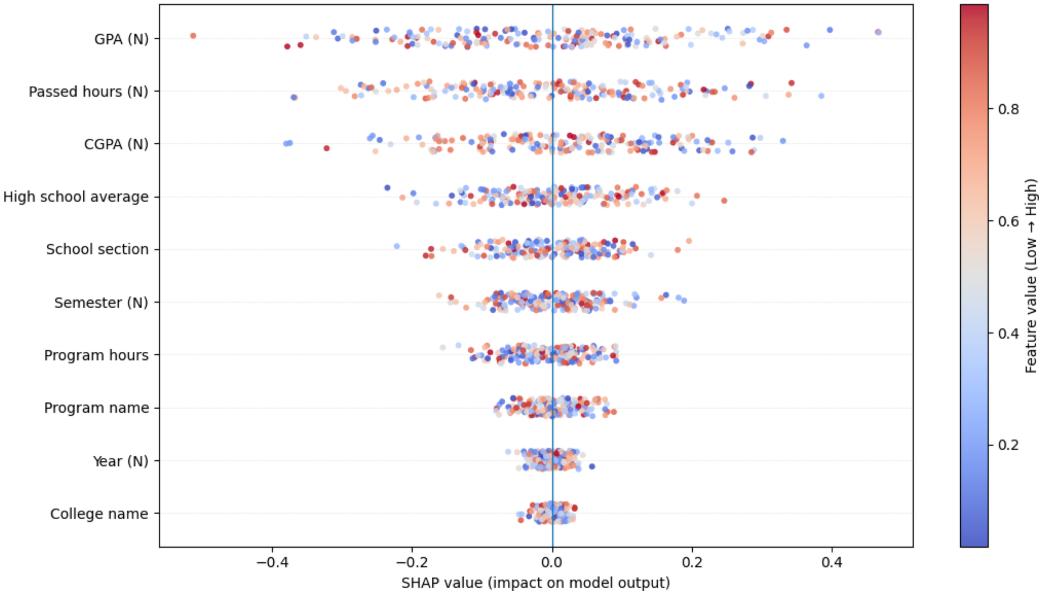


Figure 6 **SHAP summary plot.** Full-size DOI: 10.7717/peerj-cs.3792/fig-6

a clear and user-friendly picture of which characteristics have the greatest influence on the decision-making process.

The explanation results show that, the academic performance and progress indicators, especially GPA (N), Passed hours (N), and CGPA (N) are the strongest key to forecasting the possibility of getting academic warning in the future. The results of this study are consistent with the feature selection results and indicate that the model is based more on meaningful semester-level academic predictors instead of recreating institutional warning rules. Other background and contextual factors (high school average, school section and semester (N)) offer complementary predictive data needed to narrow the risk estimation but have no overshadowing effect in the model.

These observations can also be further supported by Fig. 7 which provides a global feature importance bar plot using the mean magnitude of absolute contributions. The influence of each input variable in every instance considered is summarized in this visualization, and the relative influence of each feature in predicting model output has been focused on. The findings indicate that a combination of Semester-N features is useful in predicting academic warning, which supports the soundness and readability of the suggested system.

In addition to interpretability, the SHAP-based explanations synthesize the increased utility of the system in terms of actionable recommendations. By establishing the academic indicators that have the strongest predictive power, advisors can also match interventions with the specific needs of students, including correcting declining semester GPA, tracking the lack of enough credit progress, or offering specific academic assistance. This open connection between the prediction and intervention enhances the confidence in the system and makes data-based decision-making in the academic advising processes easy.

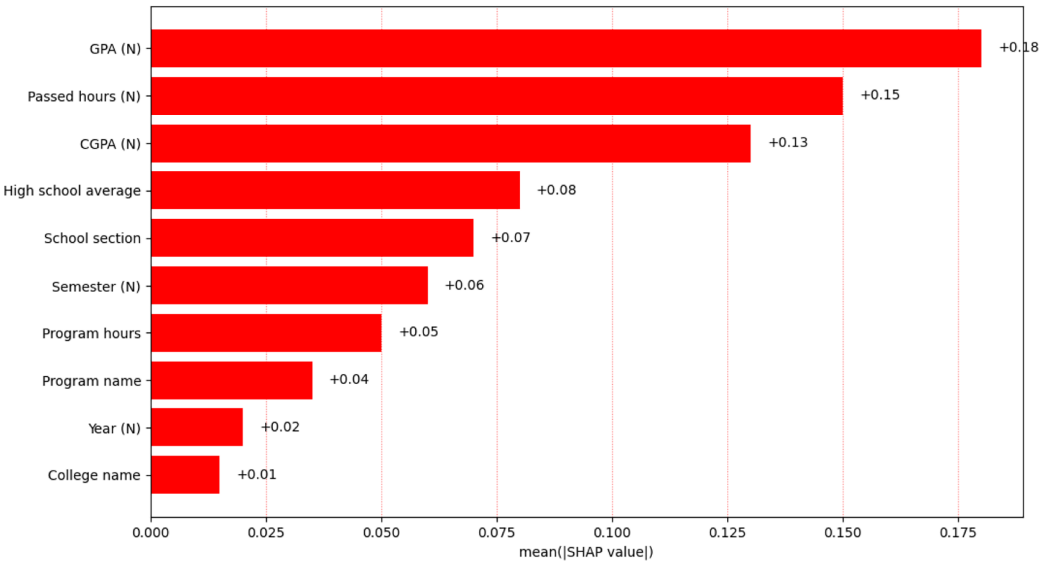


Figure 7 **SHAP-style bar plot showing global feature importance for the selected prediction model under the temporal academic warning framework (Semester N → N + 1).**
Full-size DOI: 10.7717/peerj-cs.3792/fig-7

Even though the results of the explanation show that the internal validity of the results is high, it is worth noting that the findings are based on a one-institution dataset. Various academic rules, curriculums, and pupil baseline in various institutions can have an effect on patterns of feature importance. Thus, the explainability results should be confirmed by future working with multi-institution datasets in order to further evaluate the generalizability.

These results are in line with the other published works in the area of academic performance prediction in which it was evident that ensemble-based classifiers (RF, Gradient Boosting, and XGBoost) outperform a variety of conventional models such as LR, NB, KNN. As the previous studies have revealed, ensemble methods are especially useful in the educational data mining because they can model nonlinear and complex relationships between academic and demographic features, which is also proved by the findings in the work.

This study goes beyond predictive performance and incorporates model explainability into the early warning paradigm. Although a number of the past works were characterized mostly by accuracy and recall measures, the use of SHAP-based explanations in this study offers a better understanding of the relative contribution of a single academic indicator to the result of the predictions. The outcomes of the explanation show that the recent indicators of academic performance and progression, including GPA (N), CGPA (N) and Passed hours (N), are the focal factors of the future academic risk of warning which are also in tandem with the patterns frequently illustrated in the higher education analytics literature.

Notably, the SHAP analysis proves that a combination of several Semester-N features determines academic warning prediction, and no single dominant variable is involved.

This observation supports the usefulness of interpretable machine learning models in the education context, wherein the interested parties can not only learn who is at risk, but also why the model makes a specific choice. This transparency can be used to facilitate more informed and targeted academic advising and enhance the relationship between predictive analytics and actionable intervention strategies.

Despite the high internal validity of the results, it must be admitted that the analysis is grounded on the data related to one institution. The relative differences between institutions in academic policies, curriculum, and students can have an impact on feature importance patterns and model behavior. The current study should therefore be confirmed with subsequent research involving multi-institution datasets to confirm the predictive and explainability results of the current research to further determine the external validity of the proposed system.

Practical implications for higher education institutions

The research results give a number of practical implications on academic advisors, program coordinators and higher education administrators. First, the proven efficiency of machine learning models, especially ensemble-based classifiers, allows the institutions to detect at-risk students early into the academic cycle and use data of semester N to make predictions regarding the potential academic warning of semester N + 1. This diagnosis will be favorable in terms of timely and focused interventions that can help avoid academic probation, slow graduation, or low school drop-out.

Secondly, SHAP-based explainability increases transparency within the predictive process as the academic and contextual factors involved in the vulnerability of a student are very clear. Instead of using opaque model outputs, advisors will be able to perceive the effects of indicators like semester GPA, passed credit hours accumulated, and cumulative academic performance on predicted risk. Such transparency favors evidence-based advising, enhances faith in automated decision-support systems, and minimizes confusion among academic staff.

Third, the recommendation aspect of the suggested system will convert predictive knowledge into practical advice depending on individual student profiles. Through predicting risk based on interpretable feature contributions, advisors are able to provide more consistent and personal academic support, including load modifications, academic skill building, or targeted tracking of academic progress.

Last but not least, the suggested early warning and recommendation application can be embedded in the current academic portals or advising dashboards to facilitate ongoing monitoring of students. This kind of integration can lead to less work of an advisor, greater consistency in intervention strategy, and student success outcomes within an institution. In general, the system has offered a scalable and transparent decision-support framework that aligns predictive analytics and practical academic advisory requirements in colleges or universities.

CONCLUSION

An intelligent warning system for college undergraduates was proposed through machine learning methodology in higher education institutions to evaluate undergraduate performance. The research aimed to develop precise predictions of academic warning recipients while providing data-based recommendations for proactive academic interventions. A machine learning evaluation was conducted on fifteen different traditional and ensemble and hybrid models using academic, institutional and demographic attributes from a wide dataset. The key metrics for evaluation of performance included accuracy, precision and recall and F1-score together with AUC measurements. Of all available predictive models' ensemble methods chosen specifically based on their performance amounted to Boosting with Grid Search, Stacking, XGBoost and RF which produced the most accurate results with elevated AUC scores.

The selection of features SCGPA and GPA and CGPA proved most consequential to academic performance by applying multiple evaluation techniques which included correlation-based and gain-ratio along with information gain relief and symmetrical uncertainty. The SHAP-based model explainability implementation provided essential interpretive capabilities which revealed inner mechanisms of selected best models along with understanding their dependency on individual student variables. The interpretability tools strengthened system transparency while ensuring both trust and ethical compliance of the proposed system.

The experimental results confirm that machine learning can successfully be used to develop warning systems in higher educational settings. The study provides usable evidence that helps academic advisors alongside institutional decision-makers to deploy data-powered methods which boost student retention alongside achievement levels. The system will be developed in three stages through recommendation system implementation and multi-class academic risk level extension before moving to real-time institutional deployment.

Limitations

Although the results obtained in the present study are quite promising, one has to admit several limitations. To begin with, the data used was collected in one learning institution, which can limit the extension of the results to other learning institutions, or even other nations with different curriculums, student population, or other institutional procedures. Second, the proposed recommendation system is data-driven and exhibits high rates of predictive performance; however, the system has yet to take into consideration other psychological or behavioral-based factors, *e.g.*, student motivation, attendance, or engagement, which are known to influence academic outcomes. Third, the system is not developed to be dynamically adaptive and currently does not allow real-time prediction of academic behavior of a student and adapt to it in a dynamic way reflecting the changes in the academic behavior of students over a semester or academic year.

To overcome these shortcomings, subsequent study will pursue the study of the generalizability of the system by including data from various institutions with varied student bodies. Also, the data concerning behavior and engagement, including the records

of class attendance and learning management system activity logs, will be incorporated to give a more comprehensive picture of student performance. The recommendation engine will further be expanded to provide real-time feedback and provide continuous monitoring, such that proactive interventions can be made, which can be updated as the student data shifts with time.

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Competing Interests

The authors declare that they have no competing interests.

Author Contributions

- Sameh Taqatqeh conceived and designed the experiments, performed the experiments, analyzed the data, performed the computation work, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Mary Agoyi analyzed the data, authored or reviewed drafts of the article, and approved the final draft.

Data Availability

The following information was supplied regarding data availability:

The raw data is available in the Supplemental File.

Supplemental Information

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